

RES-5 endpoint DR-2 floor bound (v1.2, grade-corrected): EXACT closure for

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This corrects the grade of v1.1. The endpoint reduces to the floor threshold $K_{\text{floor}} < 26.2$ (Section 2); the honest status against the existing DR-2 results is a three-way split (Sections 3–5), NOT the flat "unconditional" of v1.1. Scope: the $I = 2 \times 10^{-3}$ endpoint; the lattice competitor class = B1's T6 scope. No tier action.

2. The closure threshold (recalled)

With the corrected tail 0.059, the endpoint closes for a competitor Q iff

$$K_{\text{floor}}(Q) = \frac{E_+(Q)}{N^2} - 1 < 26.2, \quad E_+(Q) = \sum_t w_t^2, \quad N = |Q|.$$

3. Enumerated competitors: EXACT closure (rigorous)

For the actual Reading-H competitor list (single shells $\{100\}, \{200\}, \{111\}, \{110\}$ and multi-shell unions up to n_{pack}), the additive energy is an EXACT combinatorial count and gives

$$K_{\text{floor}}^{\text{enum}} \leq 12.0 < 26.2 \quad (\text{factor} \geq 2.2),$$

so the endpoint closes RIGOROUSLY for the enumerated competitors. This part is unconditional and exact.

4. Full lattice class: STRONG EVIDENCE via R-026 (not a theorem)

R-026 (dr2-lattice-divisor-closure) bounds the lattice additive energy,

$$E_+(Q) \leq (1 + C_\varepsilon R^\varepsilon) N^2, \quad K_{\text{floor}} \leq C_\varepsilon R^\varepsilon,$$

DECOUPLING-FREE, at grade **T7 MODULO TEXTBOOK NUMBER THEORY** (the divisor bound and Dirichlet's class-number formula). Two residuals keep this STRONG EVIDENCE rather than a theorem for the endpoint:

- the constant: $C_\varepsilon R^\varepsilon < 26.2$ over the admissible R -range is ASSERTED from the divisor estimate (the realized enumerated values ≤ 12 are a factor 2.2 below), not pinned to an explicit C_ε ;
- the link: the carrier-richness $\chi(P) \lesssim T'$ connection from E_+ to the actual STEP-5B floor is, per the B5 record, an OPERATOR-DECISION – R-026/R-027 did NOT flip DR2-SHARE (B5 stayed T5, B1 T6, H-ADM-COH retained at issue).

So the full-lattice-class endpoint closure is strong evidence (a subpolynomial, decoupling-free, T7-modulo-textbook bound plus the exact enumerated anchor), not an unconditional theorem.

5. Arbitrary- Q (not B1's scope): T6 conditional on decoupling

For an arbitrary finite $Q \subset S^2$, $E_+ \leq_\varepsilon N^{2+\varepsilon}$ is T6 PROVED CONDITIONAL on Bourgain–Demeter ℓ^2 -decoupling (the arbitrary- Q reduction is the T5 induction-on-scales). This gives $K_{\text{floor}} \leq N^\varepsilon - 1 < 26.2$ for $N \leq 41$, so the endpoint also closes for arbitrary- Q conditional on decoupling – DR-2's residual for the unrestricted / T7 goal, outside B1's lattice scope.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “v1.1 said ‘unconditional’; is it?” UPHELD against v1.1 (operator-caught): R-026 is T7 *modulo textbook NT*, and the C_ε sufficiency + the $\chi(P) \leftrightarrow T'$ link are residuals. The honest grade is EXACT (enumerated) / STRONG EVIDENCE (lattice). v1.1's "unconditional" banner contradicted its own divisor hedge and is withdrawn.
- (β) “So is the endpoint a B1 blocker?” No – the actual (enumerated) competitors close EXACTLY ($\leq 12 < 26.2$), so RES-5 is not an independent blocker. But "closed unconditionally over the whole lattice class" is NOT claimed.
- (γ) “Non-uniform amplitudes / the $\chi(P)$ link.” VALID: the weighted additive energy (R-027 bridge) and the carrier-richness link carry their own residuals; they are operator-decisions in the B5 record, retained here as such.

Quantitative sanity check (mandatory): *threshold* $K_{\text{floor}} < 26.2$; *enumerated* ≤ 12 (EXACT, factor 2.2); *lattice* $C_\varepsilon R^\varepsilon$ subpoly (T7-modulo-textbook, decoupling-free, + residual links); *arbitrary- Q* $N^\varepsilon < 27.2$ for $N \leq 41$ (T6-cond decoupling).

7. Result footer

Result ID:	B1-RH-ENUM / RES-5 endpoint DR-2 floor bound (v1.2)
Precise statement:	Endpoint closes iff $K_{\text{floor}} = E_+ / N^{2-1} < 26.2$. (i) ENUMERATED competitors: EXACT $K_{\text{floor}} \leq 12 < 26.2 \rightarrow$ closes rigorously. (ii) FULL lattice class: STRONG EVIDENCE via R-026 ($E_+ \leq (1+C_\varepsilon R^\varepsilon)N^2$, T7 MODULO TEXTBOOK NT, decoupling-free) + exact anchor; residuals = the $C_\varepsilon < 26.2$ sufficiency over the admissible range (asserted) + the $\chi(P) < T'$ link (operator-decision; R-026/R-027 did NOT flip DR2-SHARE). (iii) ARBITRARY-Q (not B1's scope): $E_+ \leq N^{2+\varepsilon}$, T6 CONDITIONAL on Bourgain-Demeter decoupling. NOT an unconditional theorem.
Scope:	$I=2e-3$ endpoint; lattice class = B1 T6 scope; uniform amplitudes (weighted = R-027 bridge, own residual).

Dependencies: res5-endpoint-2pi-bound v1.2 (threshold, corrected tail);
dr2-lattice-divisor-closure v1.2 (R-026, T7-modulo-textbook);
dr2-programme-consolidation; B5 status.json (DR2-SHARE OPEN).
Supersedes v1.0, v1.1.

Evidence grade: EXACT (enumerated K_{floor}) + STRONG EVIDENCE (lattice, R-026
T7-modulo-textbook + residual links) + INHERITED (R-022
arbitrary-Q T6-cond decoupling).

Reproduction command: python codes/vacuum/res5_dr2_kappa_bound.py

Expected output: threshold 26.2; enumerated $K_{\text{floor}} \leq 12$ (EXACT); lattice
R-026 subpoly bound (T7-modulo-textbook); 4/4 PASS.

Falsification gate: An admissible enumerated competitor with $K_{\text{floor}} > 26.2$; or
R-026's divisor bound failing over the admissible R-range.

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5/GAP-2 endpoint:
EXACT for the enumerated competitors; STRONG EVIDENCE for the
lattice class (R-026 T7-modulo-textbook); T6-cond decoupling
for arbitrary-Q. NOT an unconditional theorem (v1.1 corrected).

No-overclaim statement: The actual (enumerated) competitors close EXACTLY, so RES-5 is
not an independent B1 blocker; but "unconditionally closed over
the whole lattice class" is NOT claimed -- R-026 is T7 modulo
textbook NT, with the C_{eps} sufficiency + $\chi(P)$ link as
residuals. B1 unchanged (T6 on {H-LAYER}); deepest piece =
Prop-A/RES-1.

Next required action: Within B1's scope, H-LAYER now reduces to Prop-A/RES-1 (the
enumerated RES-5 endpoint is exact). To make the full-lattice
endpoint a theorem: pin C_{eps} $R_{\text{eps}} < 26.2$ over the admissible
range + the $\chi(P) < \sim T$ link (the B5 operator-decision items).